

File permissions in Linux

Project description

The research team at my organization needs to update the file permissions for certain files and directories within the `projects` directory. The current permissions do not reflect the level of authorization allowed for certain users. Updating these security permissions will help to keep our systems secure.

I performed the following tasks to update the current permissions.

Check file and directory details

The following demonstrates how I used Linux commands to determine the existing set of file and directory permissions for the `projects` directory.

```
researcher2@44f36c45f6e0:~$ pwd
/home/researcher2
researcher2@44f36c45f6e0:~$ ls
projects
researcher2@44f36c45f6e0:~$ cd projects
```

```
researcher2@9ec560fabale:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
researcher2@9ec560fabale:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:51 ..
-rw--w---- 1 researcher2 research_team  46 Jun 29 17:08 .project_x.
txt
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
```

The first screenshot are the list of commands that help to establish the absolute path `pwd` of the directory I was working within and to display `ls` and navigate to the directory `cd projects` I aimed to update the permissions on.

I used both the `ls -l` to display all accessible files and the `ls -la` commands to include all hidden files within the projects directory.

In the `/home/researcher2/projects` directory, there are five files with the following names and permissions:

- `project_k.txt (-rw-rw-rw-)`
- `project_m.txt (-rw-r-----)`

- `project_r.txt (-rw-rw-r--)`
- `project_t.txt (-rw-rw-r--)`
- `.project_x.txt (-rw--w----`) (hidden file)

The 10-character string in the first column represents the permissions set on each file within the directory.

Describe the permissions string

```
researcher2@9ec560fabale:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
researcher2@9ec560fabale:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:51 ..
-rw--w---- 1 researcher2 research_team  46 Jun 29 17:08 .project_x.
txt
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
```

The 10-character string can be deconstructed to determine who is authorized to access the various files within the `/home/researcher2/projects` directory.

The first characters represented for the files begin with '-' indicating these are files within a directory. The second-fourth characters indicate the read, write, and execute permissions granted to the user. The fifth-seventh characters indicate the read, write, and execute permissions granted to the group. The eighth-tenth characters indicate the read, write and execute permissions granted to the other.

In example `project_k.txt`, this is a file within a directory the first of the 10-character string is represented by '-'.
The permissions for this file are:

- user = read + write - execute (`rw-`)
- group = read + write - execute (`rw-`)
- other = read + write - execute (`rw-`)

This represents that the user, the group, and other all have the same permissions to read the file's contents and modify them. However, no permissions have 'x' in their listed, meaning that none can run this file as a program.

Change file permissions

The organization has determined that other should not have write access to any of the listed files within the `projects` directory. To adjust this, I used Linux commands to remove the current authorized access.

```
researcher2@9ec560fabale:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
researcher2@9ec560fabale:~/projects$ chmod o-w project_k.txt
researcher2@9ec560fabale:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
```

The first line of the screenshot displays the command I entered to display the current granted permissions for `project_k.txt`. The `chmod` command changes the permissions on files and directories. I entered `chmod o-w project_k.txt` to remove write permissions for 'other.' The first argument indicates what permissions should be changed with the second argument indicating on what specific file I wanted to adjust these permissions on. I used `ls -l` again to confirm that write permissions for other users under `project_k.txt` were correctly removed.

Change file permissions on a hidden file

The research team has archived **.project_x.txt**, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file. The following Linux commands demonstrate how I was able to adjust these permissions.

```
researcher2@9ec560fabale:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:51 ..
-rw--w---- 1 researcher2 research_team  46 Jun 29 17:08 .project_x.
txt
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
researcher2@9ec560fabale:~/projects$ chmod u-w,g-w,g+r .project_x.tx
t
researcher2@9ec560fabale:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 17:51 ..
-r--r----- 1 researcher2 research_team  46 Jun 29 17:08 .project_x.
txt
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
```

The first line of the screenshot displays the command used to display all files including hidden files `ls -la`. **.project_x.txt** is classified as a hidden file because it begins with a `'.'` I used `chmod u-w,g-w,g+r .project_x.txt` to remove write permissions from user and group `u-w`, `g-w` and to add read permissions to group `g+r`. I used `ls -la` again to confirm that only read permissions for user and group users under **.project_x.txt** were in place and that all write permissions were removed.

Change directory permissions

The organization has also updated their request permissions that only the `researcher2` user should be allowed to access the `drafts` directory and its contents. The following demonstrate how I was able to adjust these permissions.

```
researcher2@9ec560fabale:~$ ls
projects
researcher2@9ec560fabale:~$ cd projects
researcher2@9ec560fabale:~/projects$ ls
drafts project_k.txt project_m.txt project_r.txt project_t.txt
researcher2@9ec560fabale:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
researcher2@9ec560fabale:~/projects$ chmod g-x drafts
researcher2@9ec560fabale:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Jun 29 17:08 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 17:08 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 17:08 project_t.t
xt
```

Line 1 lists all permissions for user, group, and other for the projects directory of `/home/researcher2/projects/drafts`. The execute permissions currently allowed to group which must be removed. I used `chmod g-x drafts` to remove execute permissions from the group. Previously execute permissions were permitted but the organization reassessed this and wanted it to be removed for security purposes. I used `ls-l` again to confirm that execute permissions were properly removed from group users from `/home/researcher2/projects/drafts`.

Summary

I successfully changed multiple permissions to match the updated authorization my organization wanted within `/home/researcher2/projects` for both files and directories. By using `ls -l` and `ls -la` to check the old permissions for the directory, I was able to use the `chmod` command to adjust the permissions on files and directories to meet the new standards established by my organization.

.